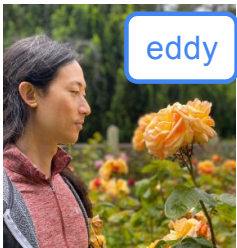




Machine Learning

Welcome!



eddy



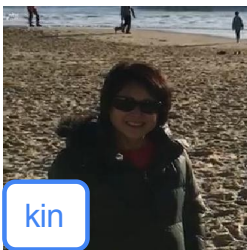
aarti



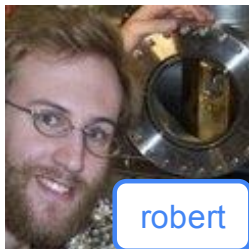
geoff



Ivy



kin



robert



andres



daniel

Re: Urgent Information :)

External

Spam x

Congratulations!
You've won
a million dollars!



Compose

Mail

Inbox

Starred

Snoozed

Sent

Drafts

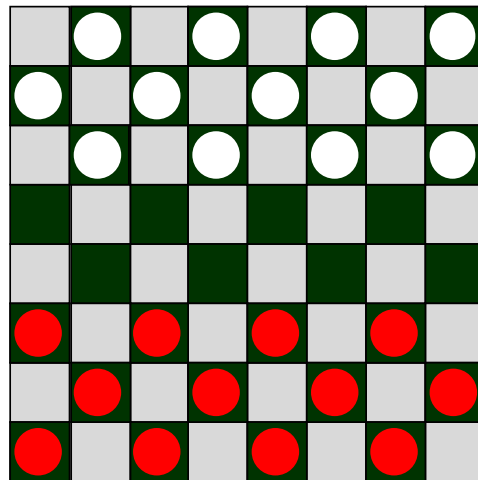
Machine Learning Overview

What is
Machine Learning?

Machine learning

“Field of study that gives computers the ability to learn without being explicitly programmed.”

Arthur Samuel (1959)



Question

If the checkers program had been allowed to play only ten games (instead of tens of thousands) against itself, a much smaller number of games, how would this have affected its performance?

- ☐ Would have made it better
- ☒ Would have made it worse

Machine learning algorithms

*rapid advancements
used most in real-world applications*

- Supervised learning
- Unsupervised learning
- Recommender systems
- Reinforcement learning

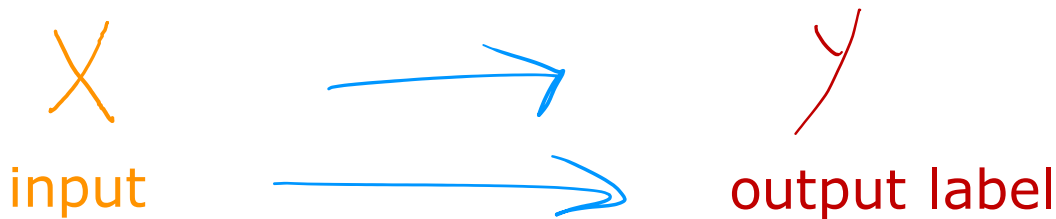
Practical advice for applying learning algorithms



Machine Learning Overview

Supervised Learning Part 1

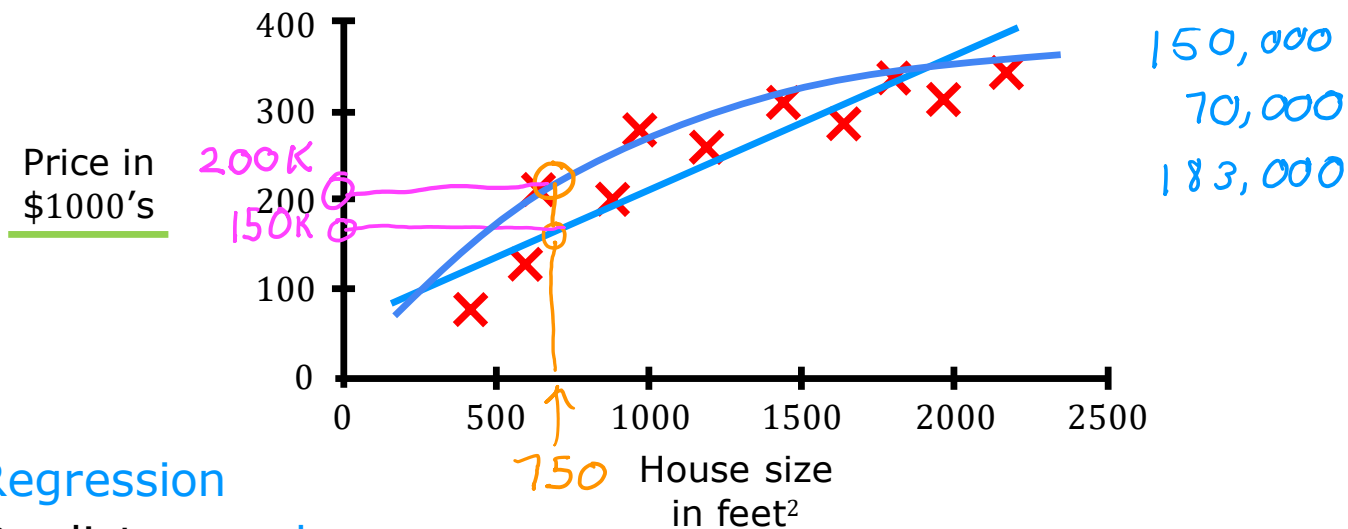
Supervised learning



Learns from being given “right answers”

Input (X)		Output (Y)	Application
email	→	spam? (0/1)	spam filtering
audio	→	text transcripts	speech recognition
English	→	Spanish	machine translation
ad, user info	→	click? (0/1)	online advertising
image, radar info	→	position of other cars	self-driving car
image of phone	→	defect? (0/1)	visual inspection

Regression: Housing price prediction



Regression

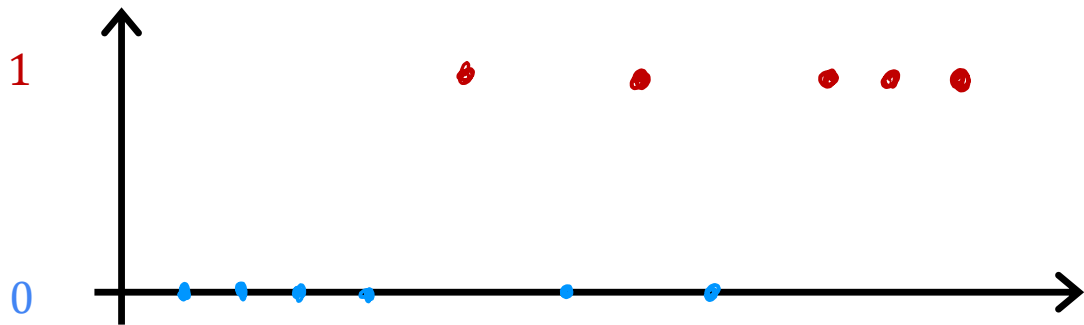
Predict a number

infinitely many possible outputs

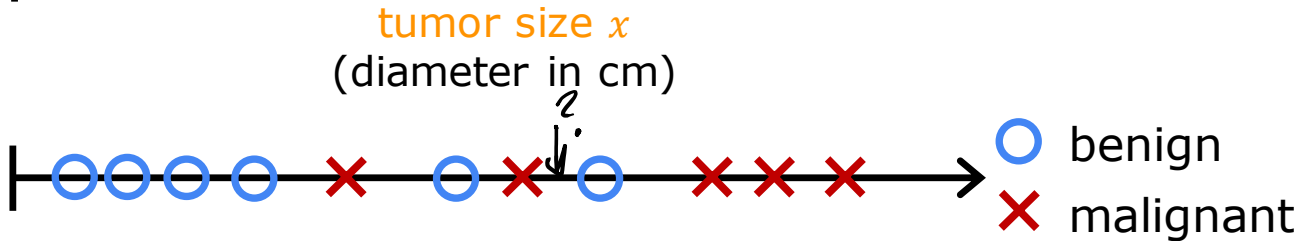
Classification: Breast cancer detection



malignant benign



size	diagnosis
2	0
5	1
1	0
7	1
⋮	



Classification: Breast cancer detection

- benign
- ✗ malignant *type 1*
- △ malignant type 2



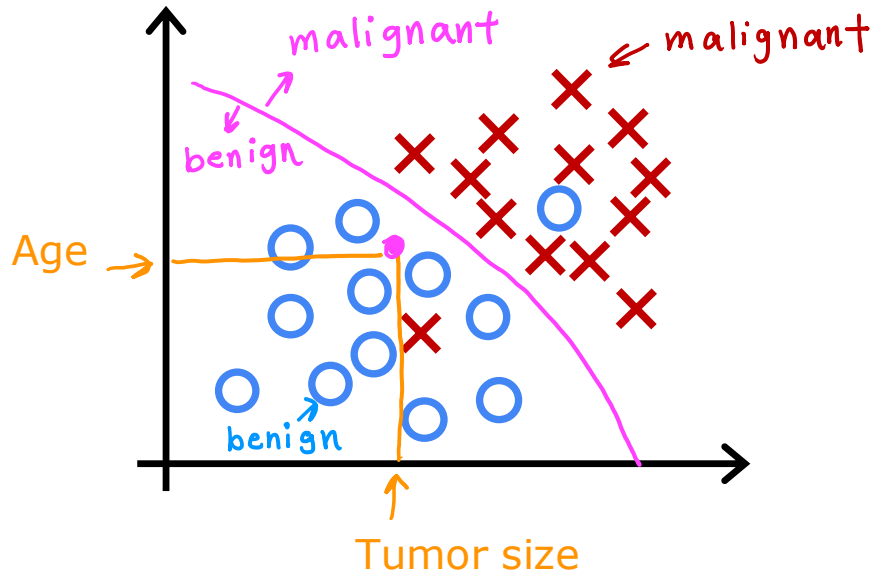
class category

Classification

predict *categories* cat dog benign malignant 0, 1, 2

small number of possible outputs

Two or more inputs



Supervised learning

Learns from being given “right answers”

Regression

Predict a number

infinitely many possible outputs

Classification

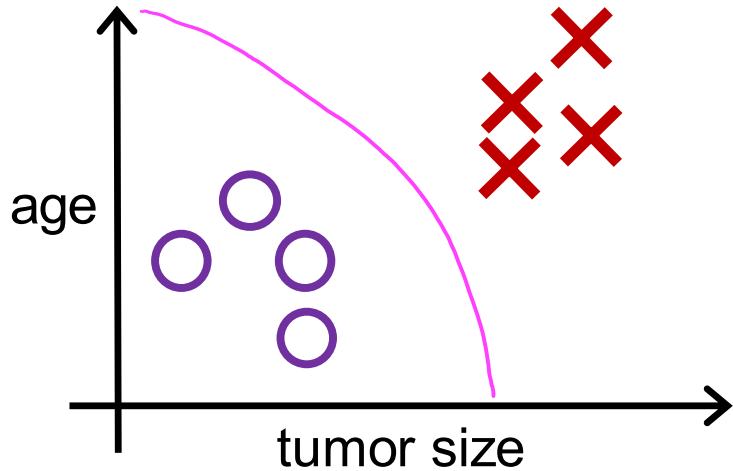
predict categories

small number of possible outputs

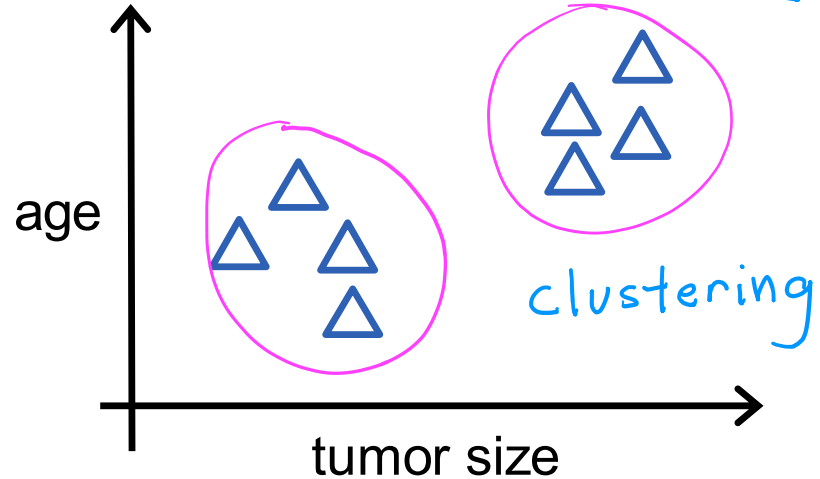
Machine Learning Overview

Unsupervised Learning Part 1

Supervised learning
Learn from data **labeled**
with the “**right answers**”



Unsupervised learning
Find something interesting
in **unlabeled** data.

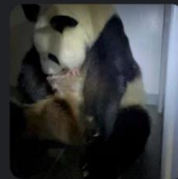


Clustering: Google news



Giant panda gives birth to rare twin cubs at Japan's oldest zoo

USA TODAY · 6 hours ago



- Giant panda gives birth to twin cubs at Japan's oldest zoo

CBS News · 7 hours ago

- Giant panda gives birth to twin cubs at Tokyo's Ueno Zoo

WHBL News · 16 hours ago

- A Joyful Surprise at Japan's Oldest Zoo: The Birth of Twin Pandas

The New York Times · 1 hour ago

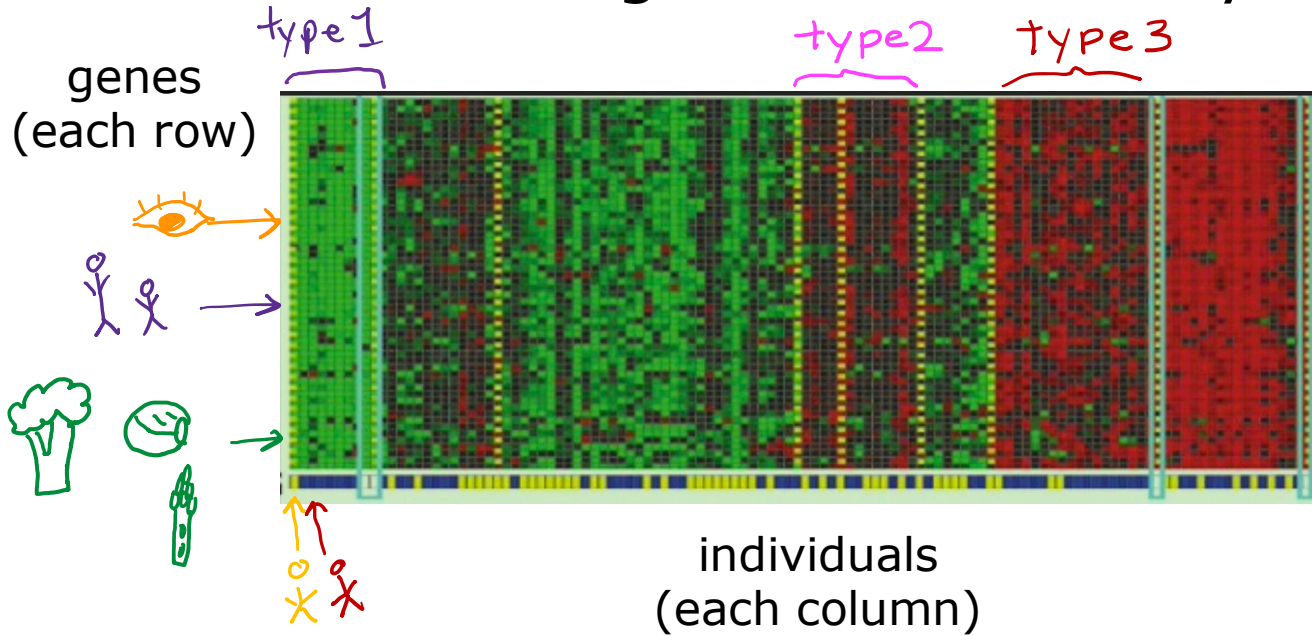
- Twin Panda Cubs Born at Tokyo's Ueno Zoo

PEOPLE · 6 hours ago

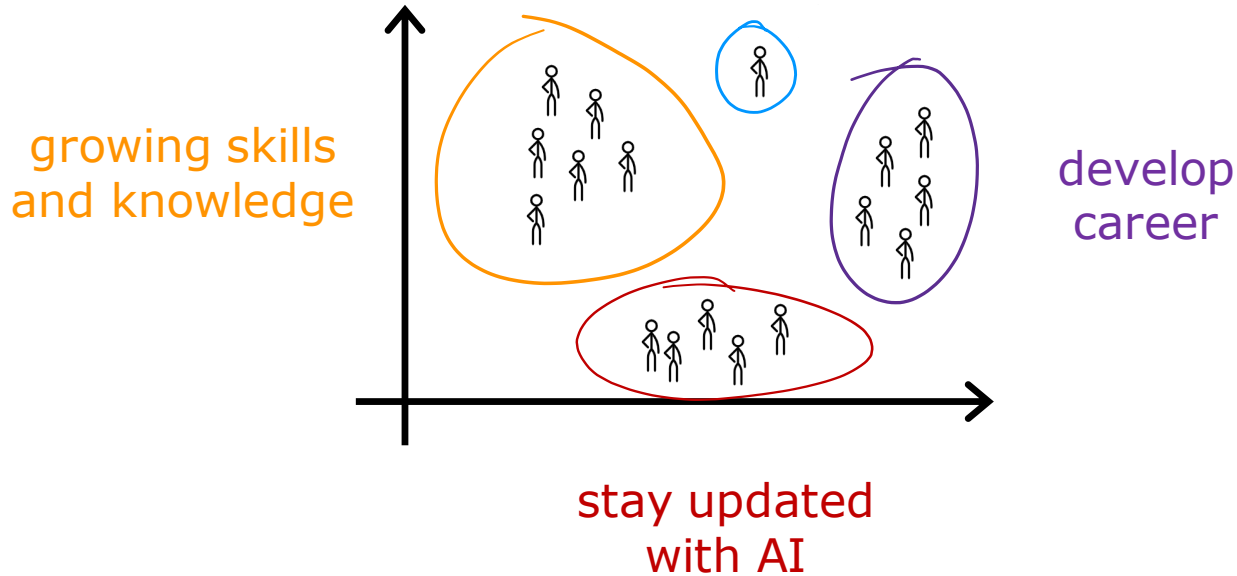
 [View Full Coverage](#)



Clustering: DNA microarray



Clustering: Grouping customers



Unsupervised learning

Data only comes with inputs x , but not output labels y .
Algorithm has to find **structure** in the data.

Clustering

Group similar data points together.

Dimensionality reduction

Compress data using fewer numbers.

Anomaly detection

Find unusual data points.

Question

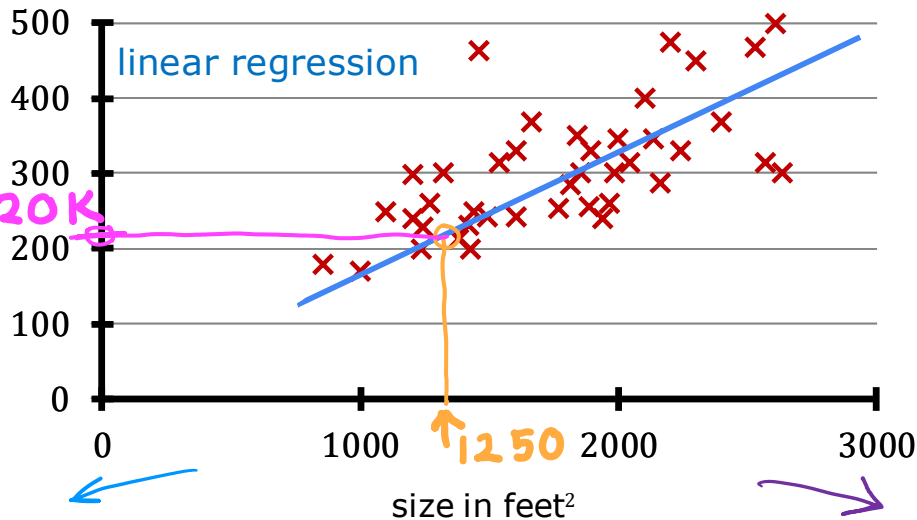
Of the following examples, which would you address using an **unsupervised** learning algorithm?

- 1 Given email labeled as spam/not spam, learn a spam filter.
- 2 Given a set of news articles found on the web, group them into sets of articles about the same story.
- 3 Given a database of customer data, automatically discover market segments and group customers into different market segments.
- 4 Given a dataset of patients diagnosed as either having diabetes or not, learn to classify new patients as having diabetes or not

Linear Regression with One Variable

Linear Regression Model Part 1

House sizes and prices



price in
\$1000's
Numbers

1.5
-33.2

categories

cat } 2
dog }

disease  10

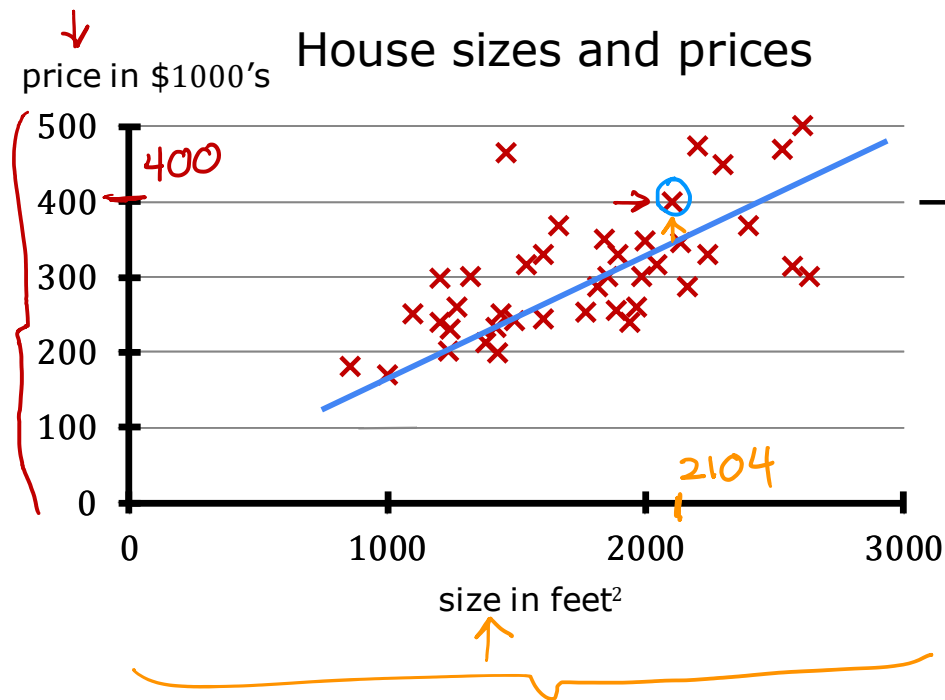
Regression model
Predicts numbers
Infinitely many possible outputs

Supervised learning model
Data has "right answers"

Classification model
Predicts categories
Small number of possible outputs

House sizes and prices

Data table



size in feet ²	price in \$1000's
2104	400
1416	232
1534	315
852	178
...	...
3210	870

Terminology

Training Data used to train the model

set: x y
→ size in feet² → price in \$1000's

(1)	2104	400
(2)	1416	232
(3)	1534	315
(4)	852	178
...	...	...
(47)	3210	870

$m = 47$

$$x^{(1)} = 2104 \quad y^{(1)} = 400$$

$$(x^{(1)}, y^{(1)}) = (2104, 400)$$

$$x^{(2)} = 1416 \quad x^{(2)} \neq x^2 \text{ not exponent}$$

Notation:

x = "input" variable
feature

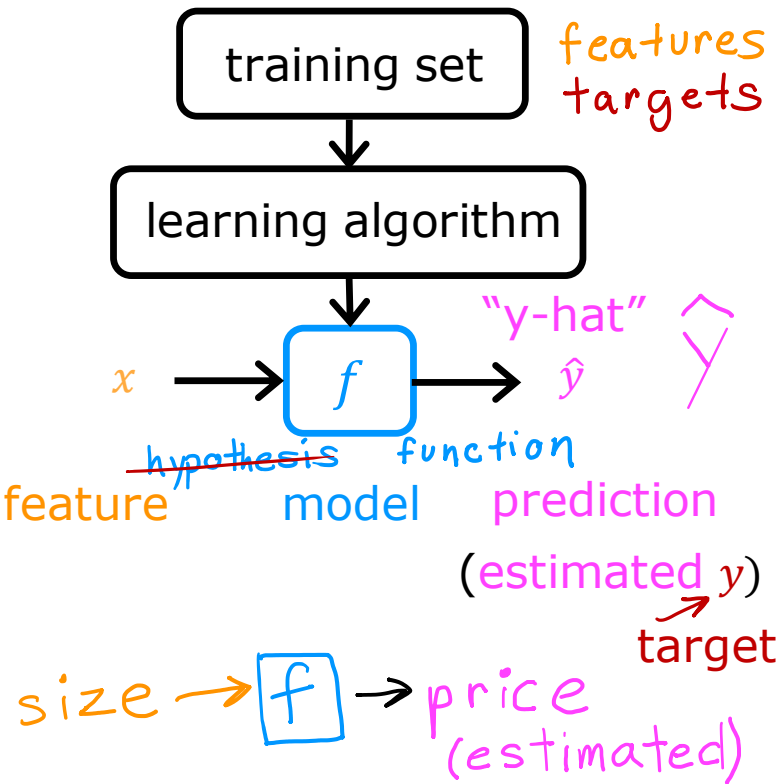
y = "output" variable
"target" variable

m = number of training examples

(x, y) = single training example

$$(x^{(i)}, y^{(i)})$$

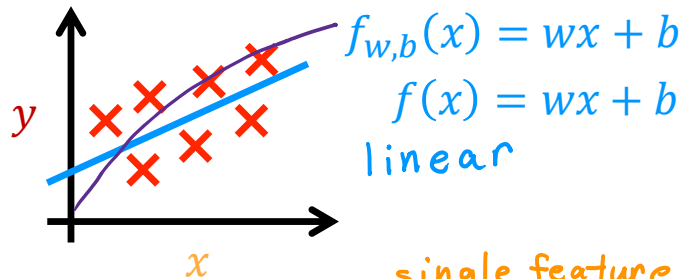
$(x^{(i)}, y^{(i)})$ = i^{th} training example
index (1st, 2nd, 3rd ...)



How to represent f ?

$$f_{w,b}(x) = wx + b$$

$f(x)$



Linear regression with *one* variable.
size

Univariate linear regression.
one variable

Linear Regression with One Variable

Cost Function

Training set

<i>features</i> size in feet ² (x)	<i>targets</i> price \$1000's (y)
2104	460
1416	232
1534	315
852	178
...	...

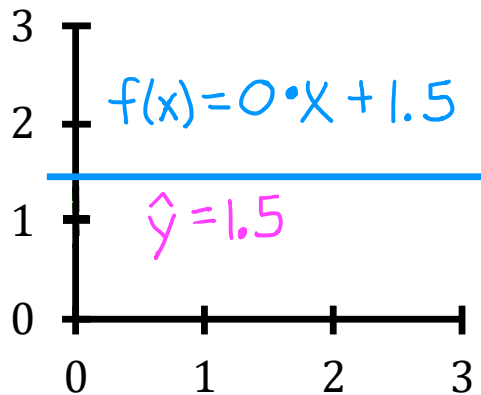
Model: $f_{w,b}(x) = wx + b$

w, b : parameters
coefficients
weights

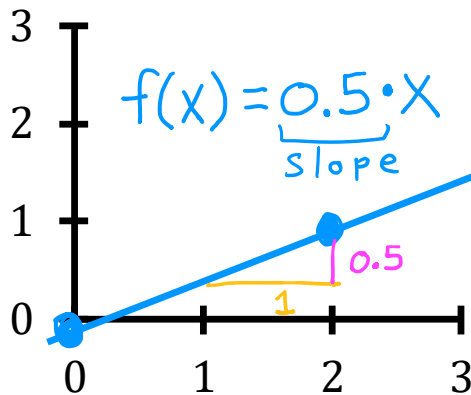
What do w, b do?

$$\underline{f_{w,b}}(x) = wx + b$$

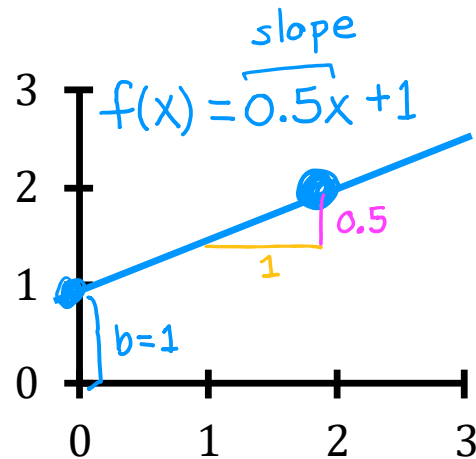
$f(x)$



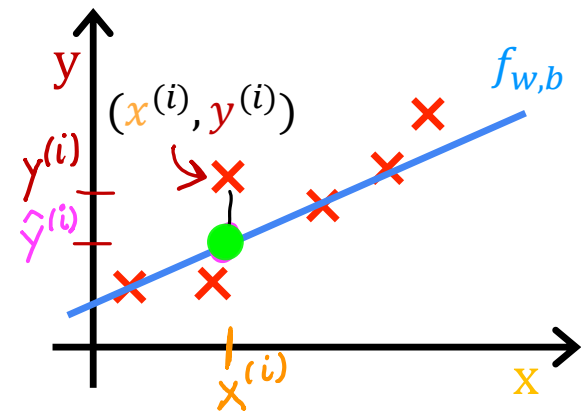
→ $w = 0$
 → $b = 1.5$
 (y-intercept)



→ $w = 0.5$
 → $b = 0$



→ $w = 0.5$
 → $b = 1$



$$\hat{y}^{(i)} = f_{w,b}(x^{(i)}) \quad \leftarrow$$

$$f_{w,b}(x^{(i)}) = wx^{(i)} + b$$

Cost function: Squared error cost function

$$J(w,b) = \frac{1}{2m} \sum_{i=1}^m \left(\underset{\substack{\text{error}}}{\hat{y}^{(i)}} - y^{(i)} \right)^2$$

m = number of training examples

$$J(w,b) = \frac{1}{2m} \sum_{i=1}^m \left(\underset{\substack{\text{intuition (next!)}}}{f_{w,b}}(x^{(i)}) - y^{(i)} \right)^2$$

Find w, b :

$\hat{y}^{(i)}$ is close to $y^{(i)}$ for all $(x^{(i)}, y^{(i)})$.

Linear Regression with One Variable

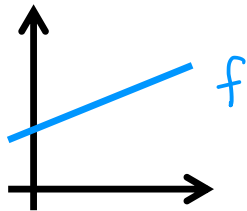
Cost Function
Intuition

model:

$$\underline{f_{w,b}(x) = wx + b}$$

parameters:

$$\underline{w, b}$$



cost function:

$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2$$

goal:

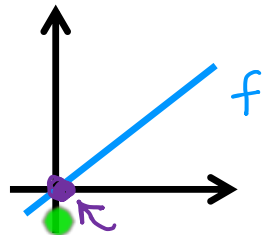
$$\underset{w, b}{\text{minimize}} J(w, b)$$

simplified

$$f_w(x) = \underline{wx}$$

$$b = \emptyset$$

w



$$\underline{J(w)} = \frac{1}{2m} \sum_{i=1}^m \underline{(f_w(x^{(i)}) - y^{(i)})^2}$$

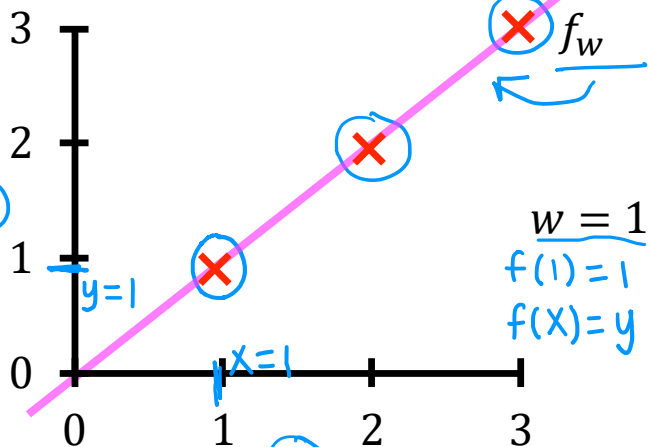
$$\swarrow w x^{(i)}$$

$$\underset{\underline{w}}{\text{minimize}} \underline{J(w)}$$

→ $f_w(x)$

(for fixed w , function of x)

input



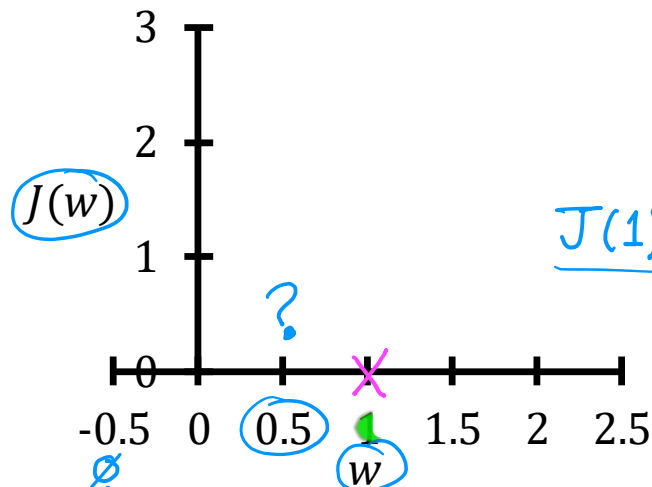
$w=1$

$$J(w) = \frac{1}{2m} \sum_{i=1}^m (f_w(x^{(i)}) - y^{(i)})^2 = \frac{1}{2m} \sum_{i=1}^m (wx^{(i)} - y^{(i)})^2$$

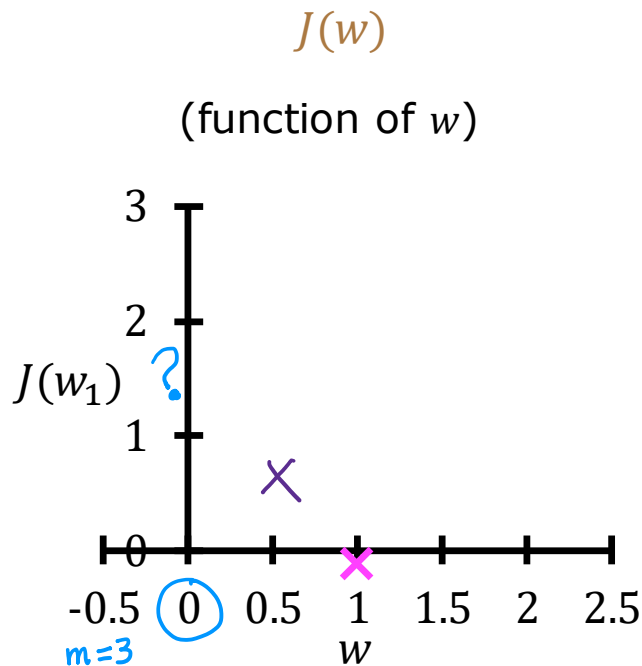
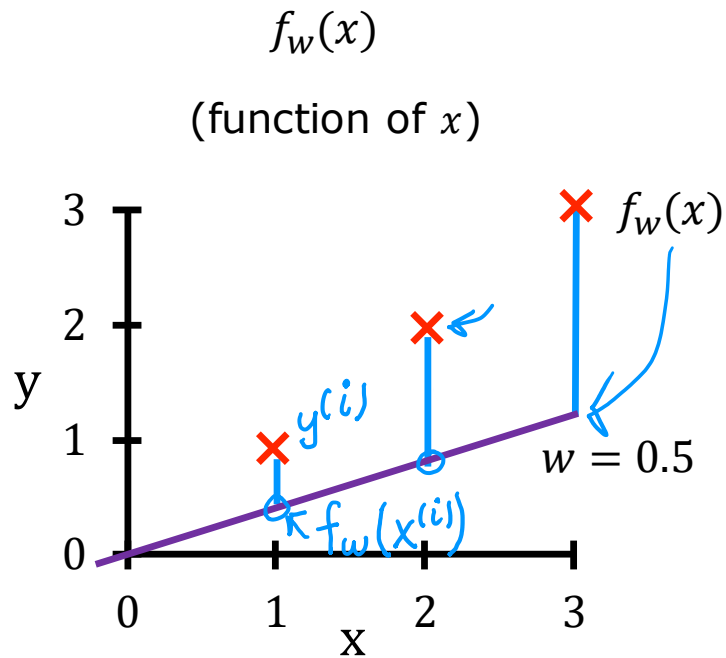
$J(w)$

(function of w)

parameter

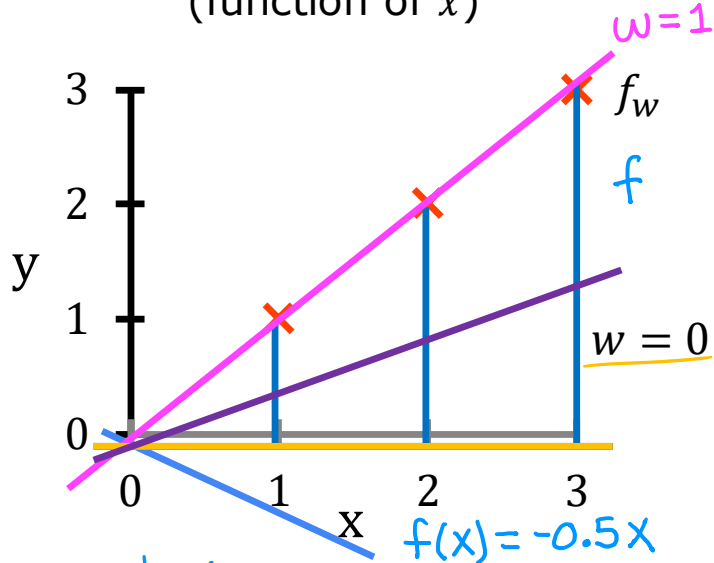


$$= \frac{1}{2m} (0^2 + 0^2 + 0^2) = 0$$

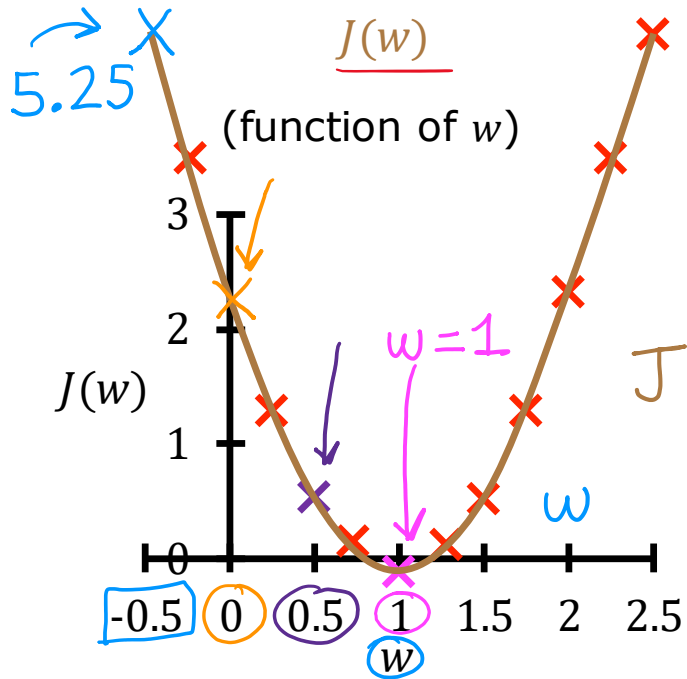


$$J(0.5) = \frac{1}{\underline{2m}} \left[(0.5-1)^2 + (1-2)^2 + (1.5-3)^2 \right] \stackrel{m=3}{=} \frac{1}{\underline{2 \times 3}} [3.5] = \frac{3.5}{6} \approx 0.58$$

$f_w(x)$
(function of x)



$$J(0) = \frac{1}{2m} (1^2 + 2^2 + 3^2) = \frac{1}{6} [14] \approx 2.3$$



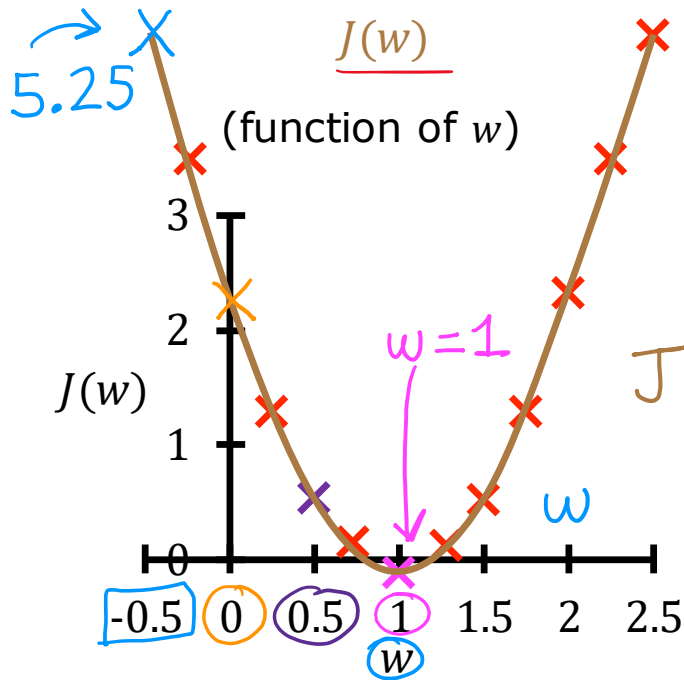
how to choose w ?

goal of linear regression:

$$\underset{w}{\text{minimize}} J(w)$$

general case:

$$\underset{w,b}{\text{minimize}} J(w, b)$$



choose w to minimize $J(w)$

Linear Regression with One Variable

Visualizing
the Cost Function

Model

$$f_{w,b}(x) = wx + b$$

Parameters

w, b ~~before: $b=0$~~

Cost Function

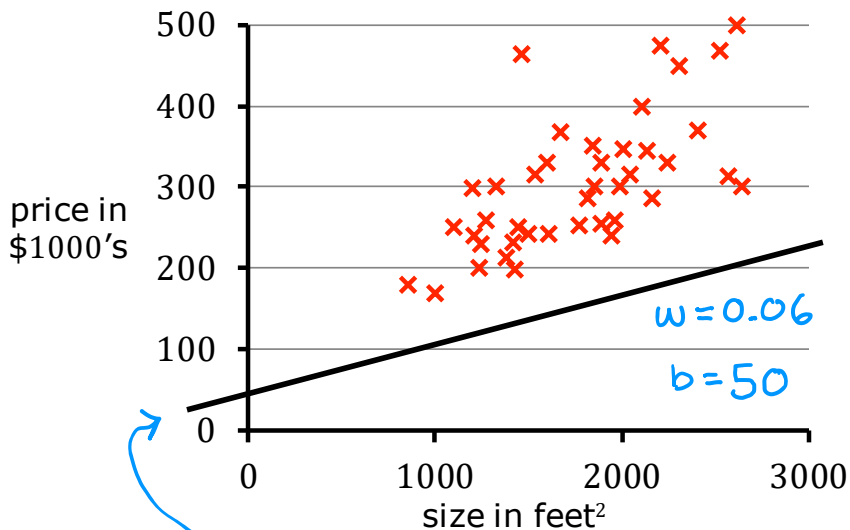
$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2$$

Objective

$$\underset{w,b}{\text{minimize}} J(w, b)$$

$f_{w,b}$

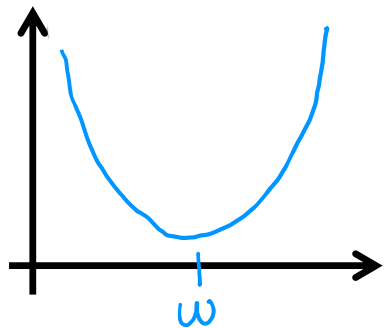
(function of x)



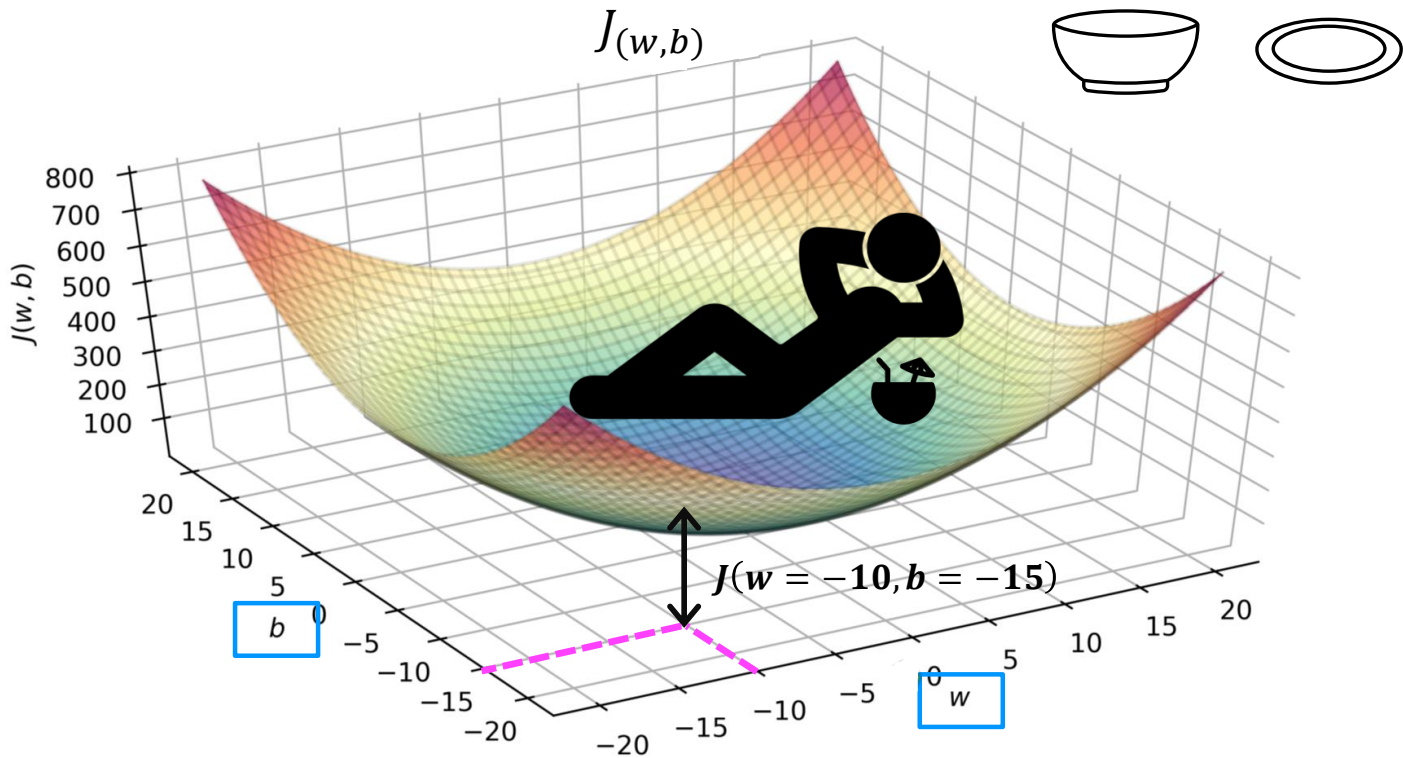
$$f_{w,b}(x) = 0.06x + 50$$

J

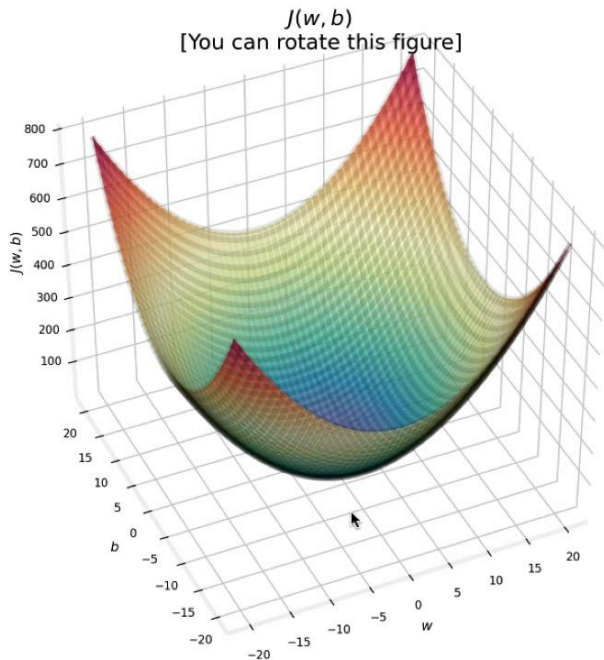
(function of w, b)



w, b



3D surface plot



Alternative
contour plot

Mount Fuji

Legend

- Contour
- Mount Fuji

Mount Fuji

Google Earth

Image Landsat / Copernicus
Data SIO, NOAA, U.S. Navy, NGA, GEBCO
Data Japan Hydrographic Association

3000 ft



Mount Fuji

Legend

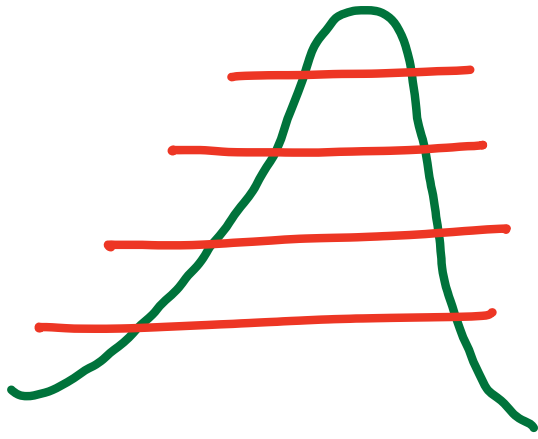
- Contour
- Mount Fuji

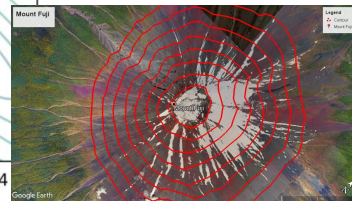
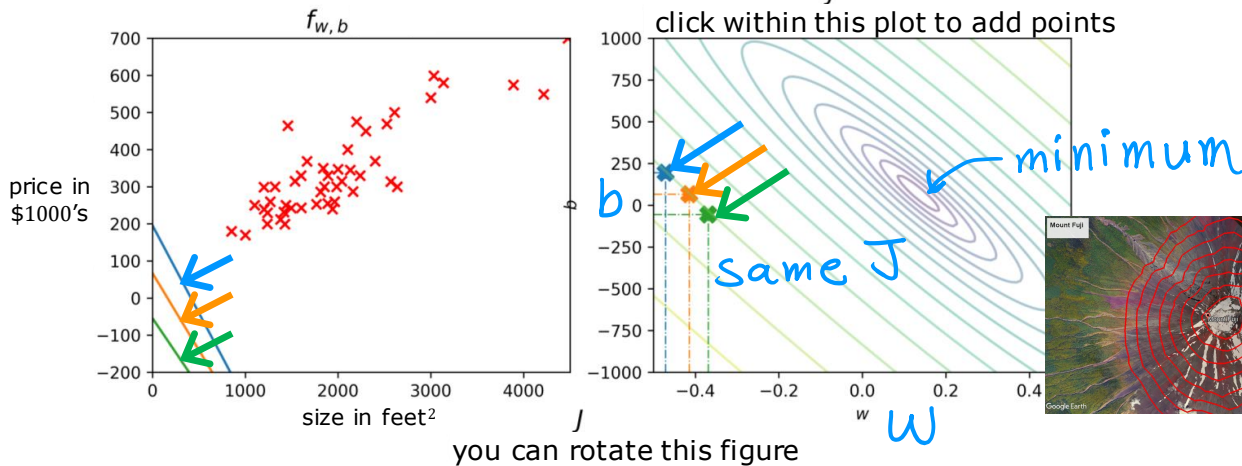
Google Earth

Mount Fuji

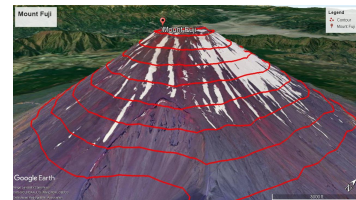
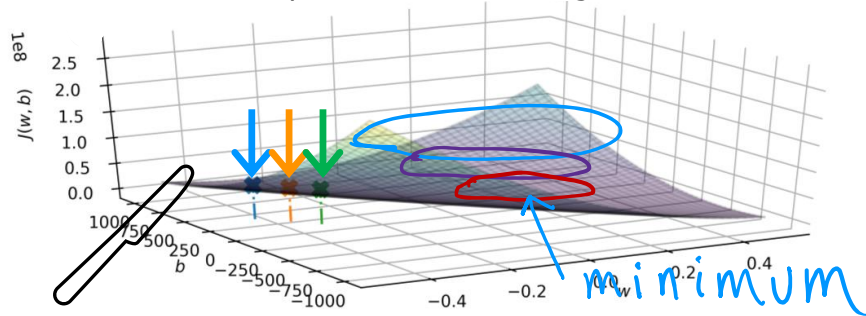
1 mi





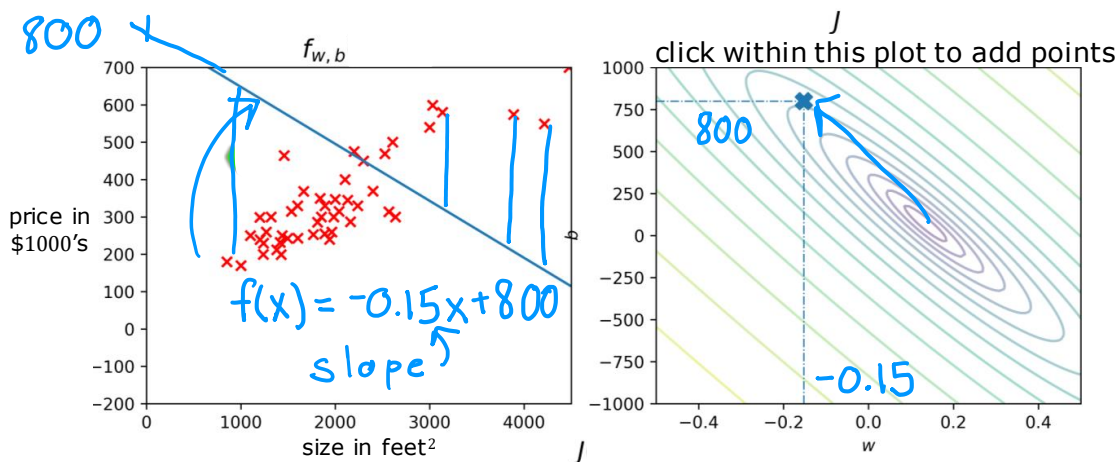


you can rotate this figure

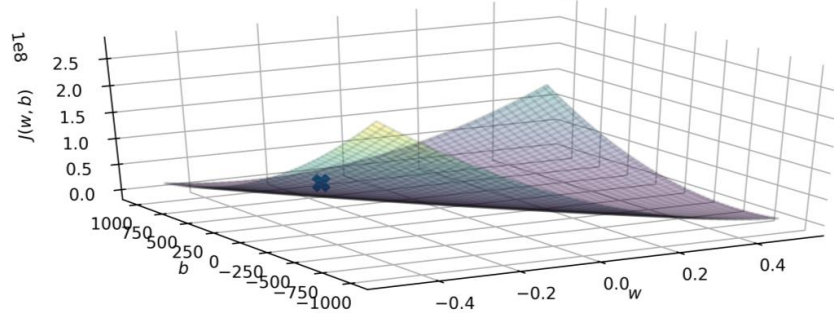


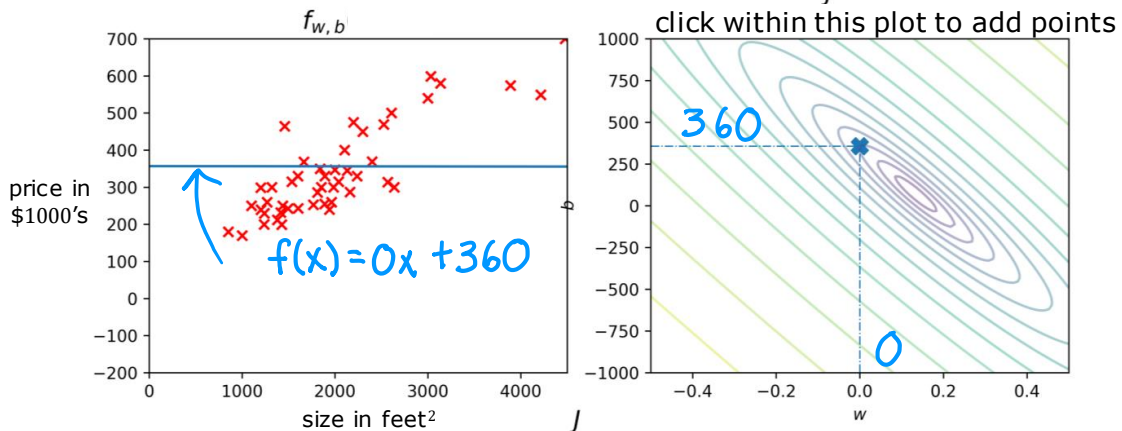
Linear Regression with One Variable

Visualization examples

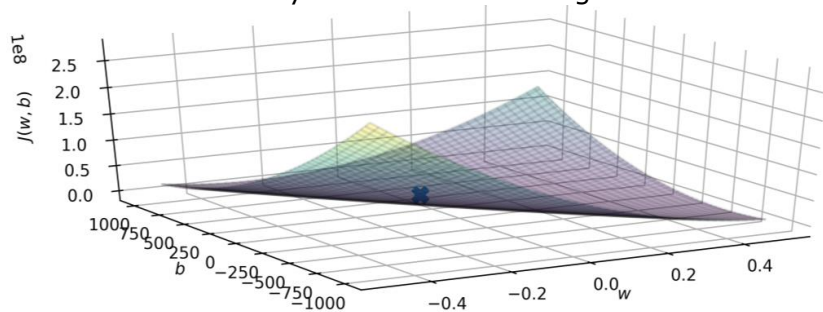


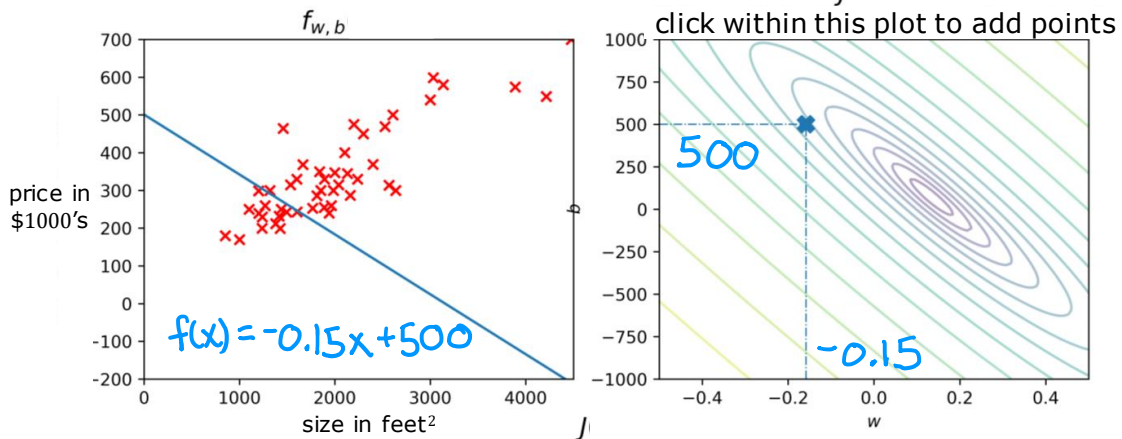
you can rotate this figure



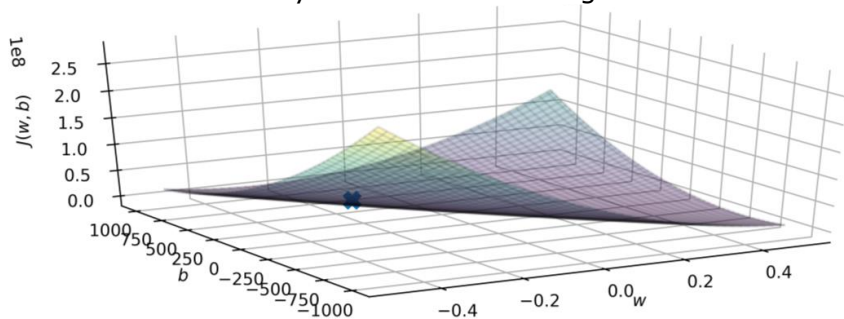


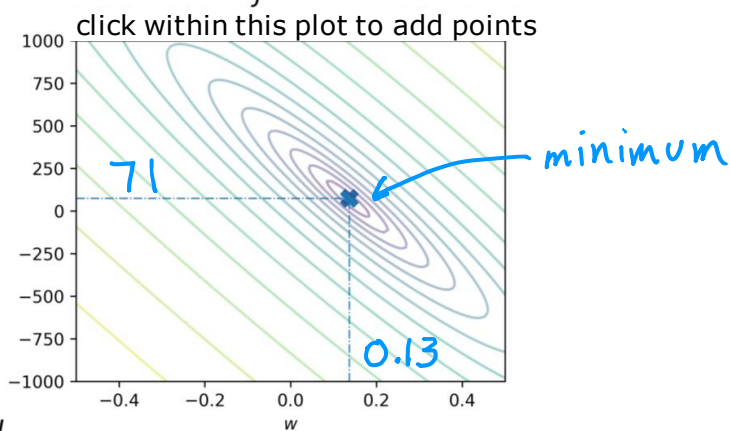
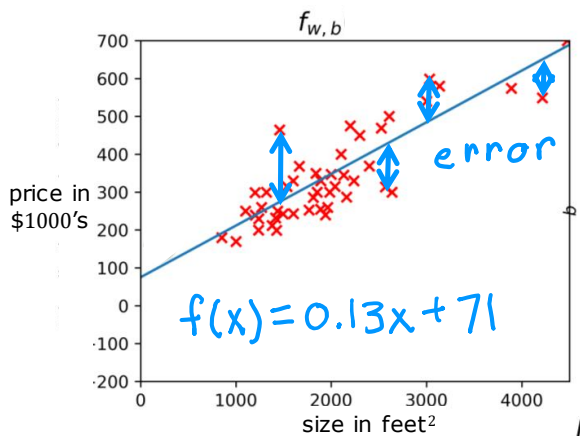
you can rotate this figure



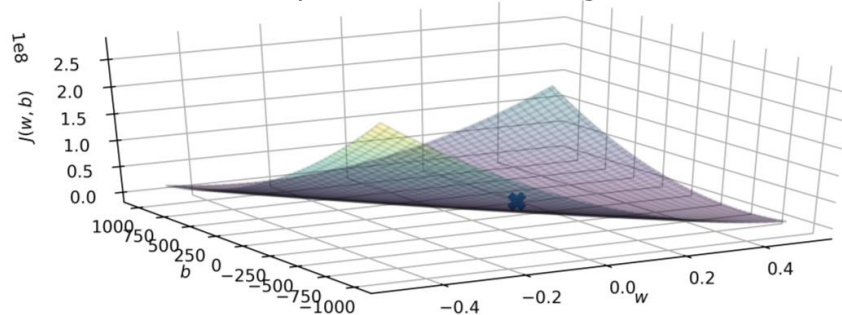


you can rotate this figure





you can rotate this figure



Training Linear Regression

Gradient Descent

Have some function $J(w, b)$ *for linear regression
or any function*

Want $\min_{w, b} J(w, b)$ $\min_{w_1, \dots, w_n, b} J(w_1, w_2, \dots, w_n, b)$

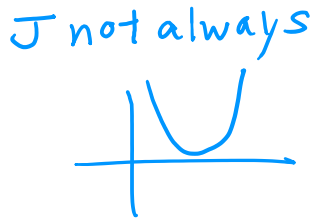
Outline:

Start with some w, b (set $w=0, b=0$)

Keep changing w, b to reduce $J(w, b)$

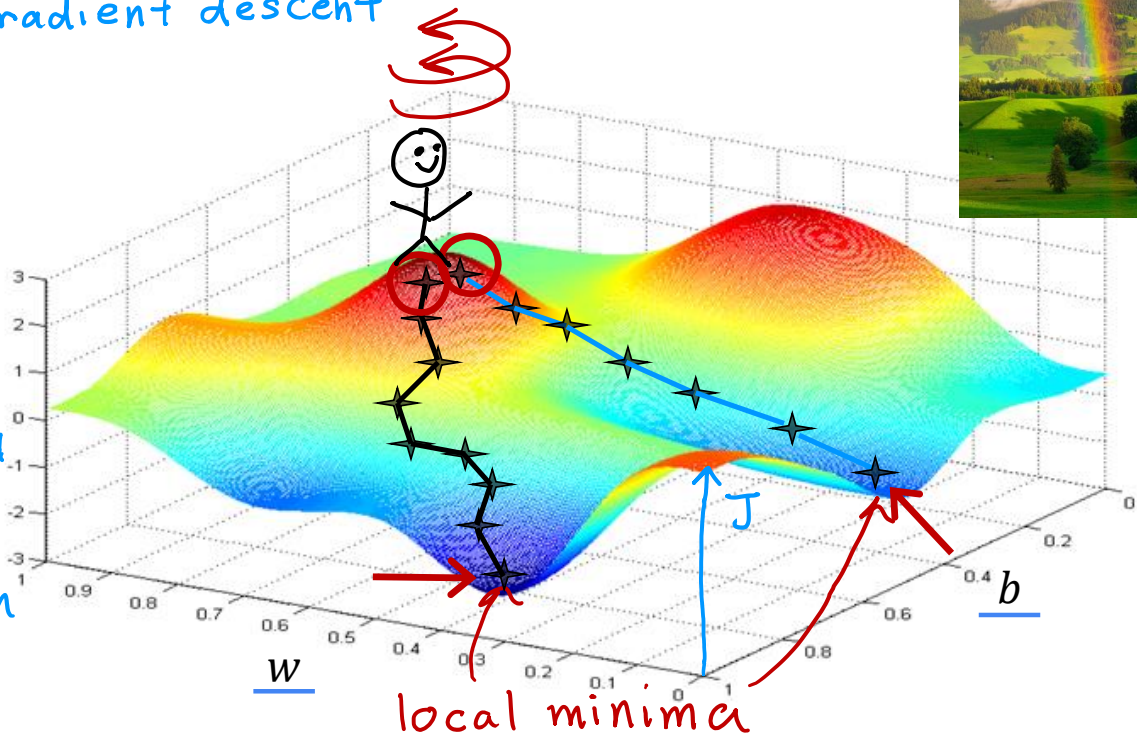
Until we settle at or near a minimum

may have >1 minimum



gradient descent

$J(w, b)$
not squared
error cost
not linear
regression



Training Linear Regression

Implementing
Gradient Descent

Gradient descent algorithm

Repeat until convergence

$$\left\{ \begin{array}{l} \underline{w} = w - \alpha \frac{\partial}{\partial w} J(w, b) \\ \underline{b} = b - \alpha \frac{\partial}{\partial b} J(w, b) \end{array} \right.$$

Learning rate
Derivative

Simultaneously
update w and b

Assignment

$$a = c$$

$$a = a + 1$$

Code

Truth assertion

$$a = c$$

$$a = a + 1$$

Math

$$a == c$$

Correct: Simultaneous update

$$tmp_w = w - \alpha \frac{\partial}{\partial w} J(w, b)$$

$$tmp_b = b - \alpha \frac{\partial}{\partial b} J(w, b)$$

$$w = tmp_w$$

$$b = tmp_b$$

Incorrect

$$tmp_w = w - \alpha \frac{\partial}{\partial w} J(w, b)$$

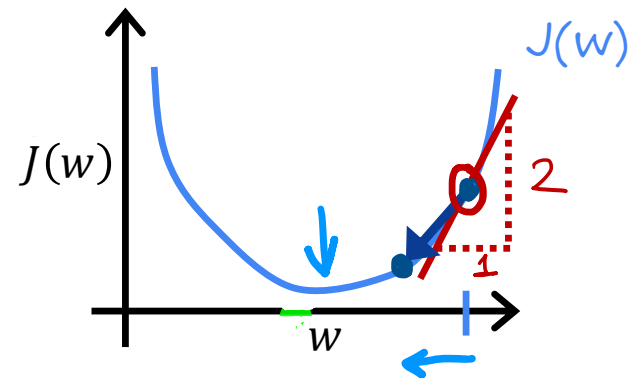
$$w = tmp_w$$

$$tmp_b = b - \alpha \frac{\partial}{\partial b} J(w, b)$$

$$b = tmp_b$$

Training Linear Regression

Gradient Descent
Intuition



$$w = w - \alpha \frac{d}{dw} J(w)$$

> 0

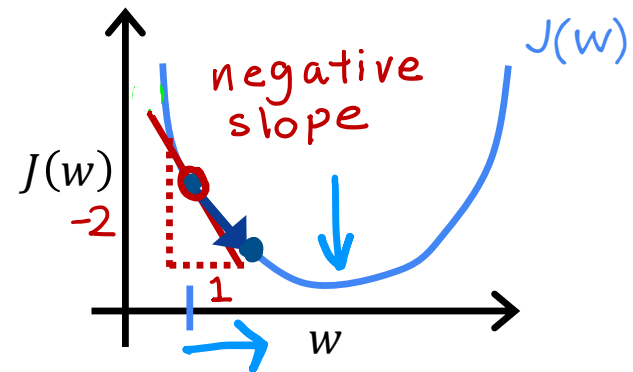
$$w = w - \underline{\alpha} \cdot (\text{positive number})$$

$$\frac{d}{dw} J(w)$$

< 0

$$w = w - \alpha \cdot (\text{negative number})$$

↑ ↑



Training Linear Regression

Learning Rate

$$w = w - \alpha \frac{d}{dw} J(w)$$

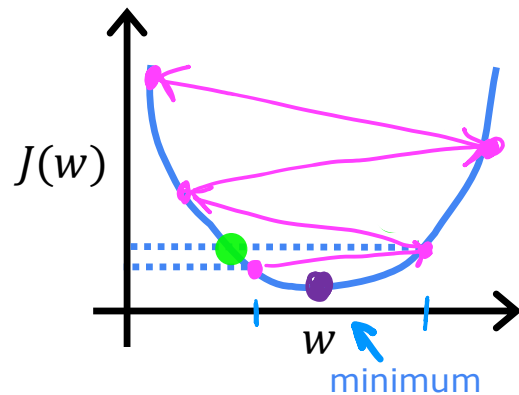
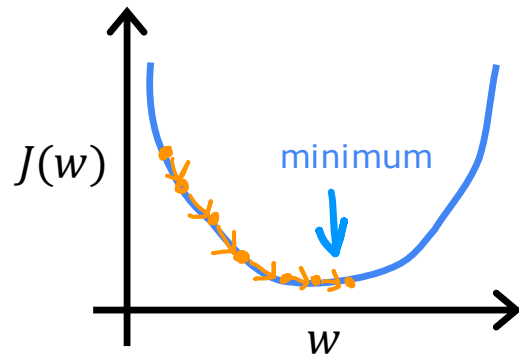
If α is too small...

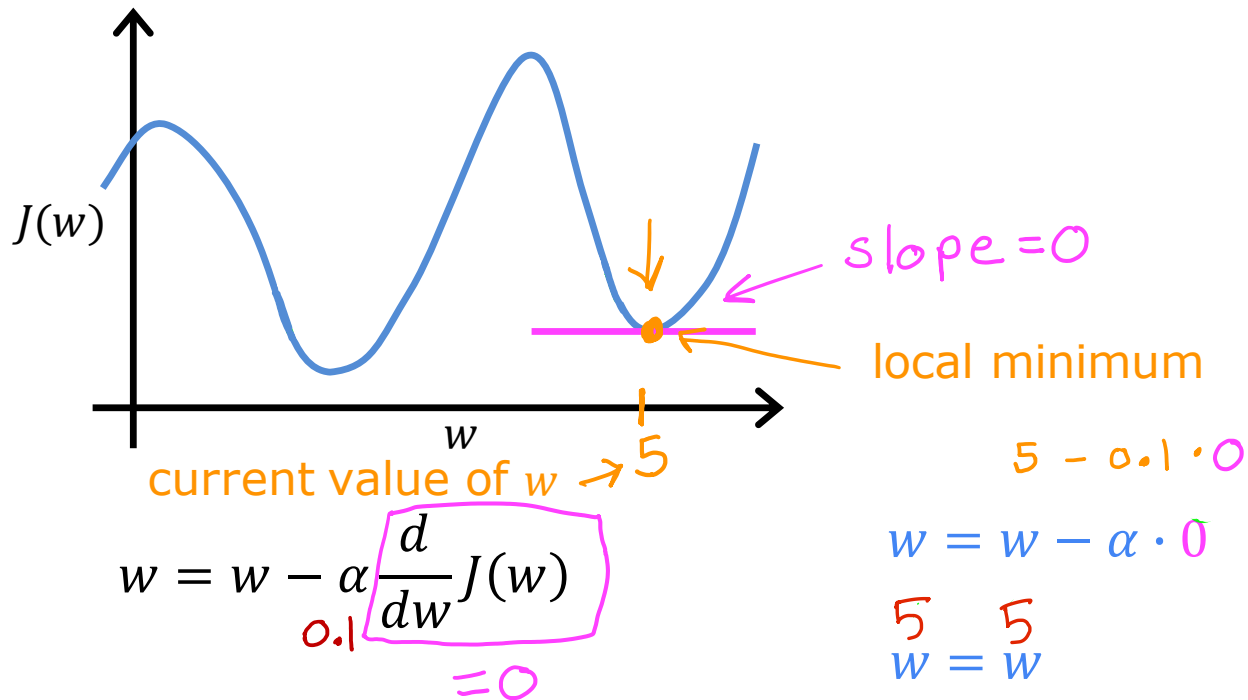
Gradient descent may be slow.

If α is too large...

Gradient descent may:

- Overshoot, never reach minimum
- Fail to converge, diverge





Can reach local minimum with fixed learning rate

α

smaller

not as large

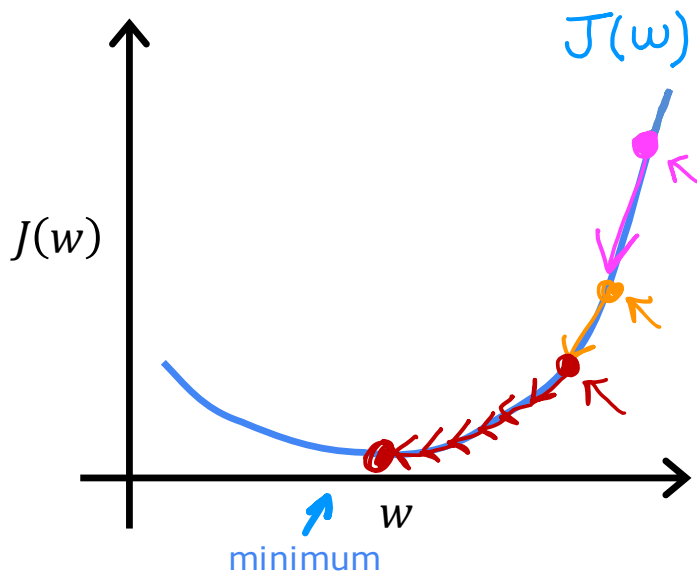
large

$$w = w - \alpha \frac{d}{dw} J(w)$$

Near a local minimum,

- Derivative becomes smaller
- Update steps become smaller

Can reach minimum without decreasing learning rate α



Training Linear Regression

Gradient Descent
for Linear Regression

Linear regression model

$$f_{w,b}(x) = wx + b$$

Cost function

$$J(w, b) = \frac{1}{2m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2$$

$(wx + b - y)$

Gradient descent algorithm

repeat until convergence {

$$w = w - \alpha \frac{\partial}{\partial w} J(w, b) \rightarrow \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)}) x^{(i)}$$

$$b = b - \alpha \frac{\partial}{\partial b} J(w, b) \rightarrow \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})$$

}

next slide
is optional!

(Optional)

$$\frac{\partial}{\partial w} J(w, b) = \frac{\partial}{\partial w} \frac{1}{2m} \sum_{i=1}^m (\underline{f_{w,b}(x^{(i)})} - y^{(i)})^2 = \frac{\partial}{\partial w} \frac{1}{2m} \sum_{i=1}^m (\underline{wx^{(i)} + b} - y^{(i)})^2$$

$$= \cancel{\frac{1}{2m} \sum_{i=1}^m (wx^{(i)} + b - y^{(i)}) 2x^{(i)}} = \boxed{\frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})x^{(i)}}$$

$$\frac{\partial}{\partial b} J(w, b) = \frac{\partial}{\partial b} \frac{1}{2m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})^2 = \frac{\partial}{\partial b} \frac{1}{2m} \sum_{i=1}^m (\underline{wx^{(i)} + b} - y^{(i)})^2$$

$$= \cancel{\frac{1}{2m} \sum_{i=1}^m (wx^{(i)} + b - y^{(i)}) 2} = \boxed{\frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})}$$

no $x^{(i)}$

Gradient descent algorithm

$$\frac{\partial}{\partial w} J(w, b)$$

repeat until convergence {

$$w = w - \alpha \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)}) x^{(i)}$$

$$b = b - \alpha \frac{1}{m} \sum_{i=1}^m (f_{w,b}(x^{(i)}) - y^{(i)})$$

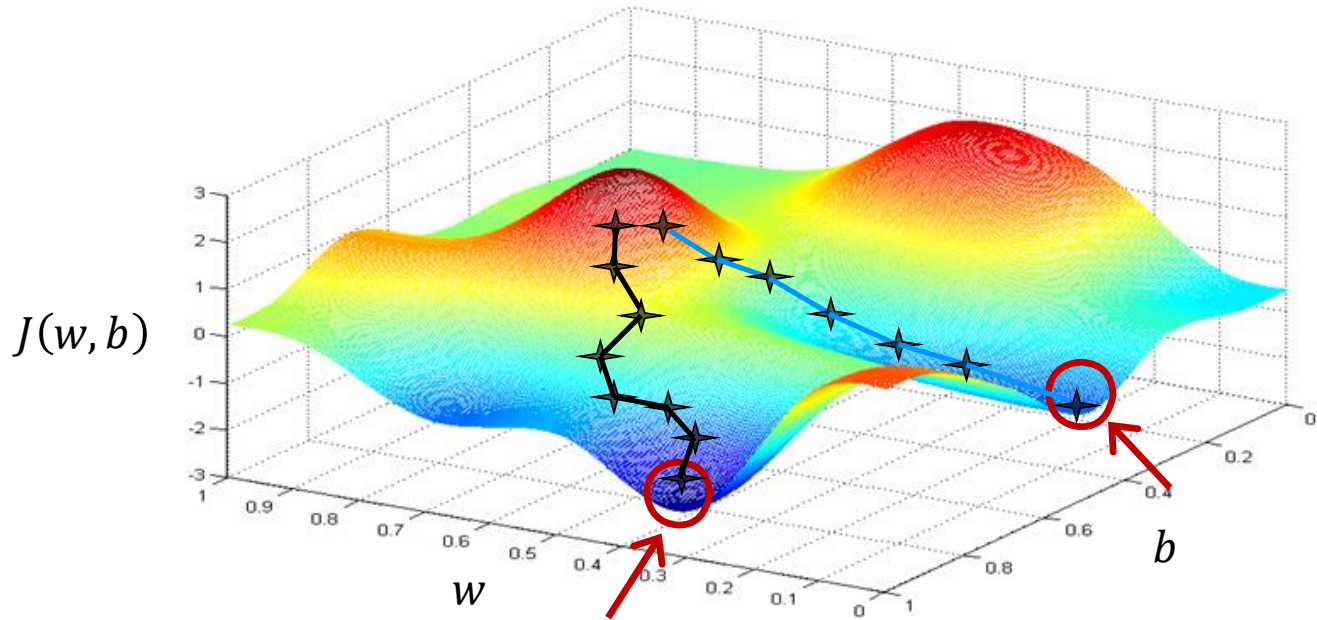
}

$$\frac{\partial}{\partial b} J(w, b)$$

Update
w and b
simultaneously

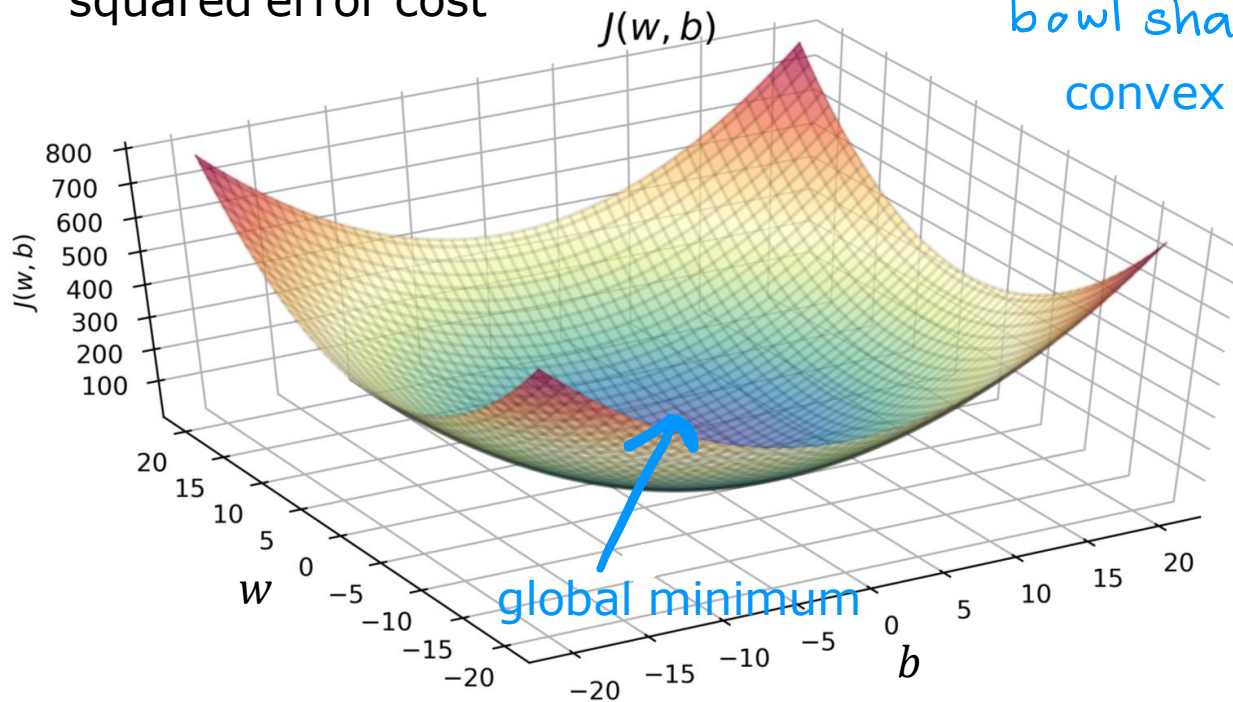
$$f_{w,b}(x^{(i)}) = wx^{(i)} + b$$

More than one local minimum



squared error cost

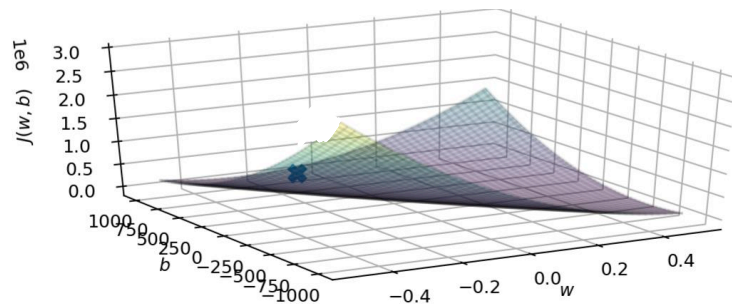
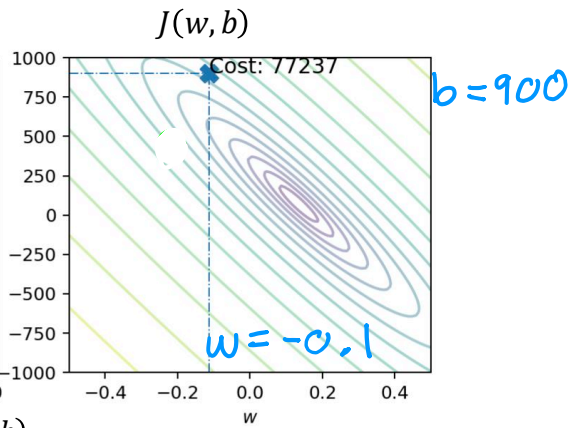
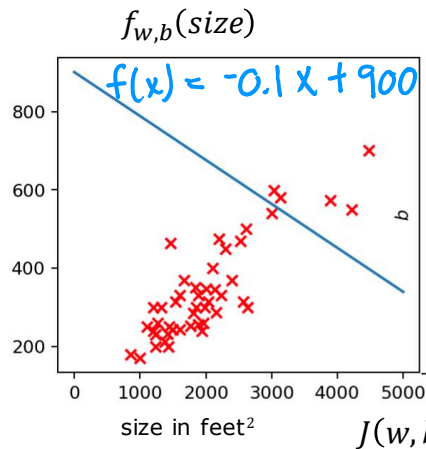
bowl shape 
convex function



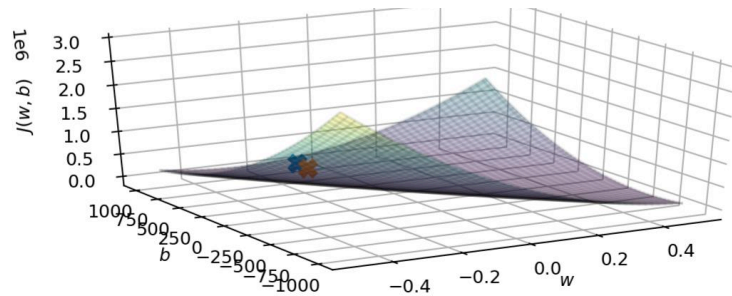
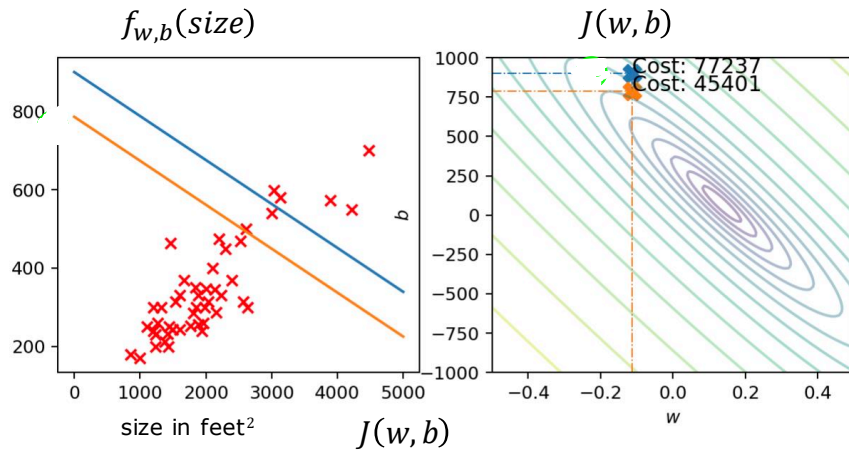
Training Linear Regression

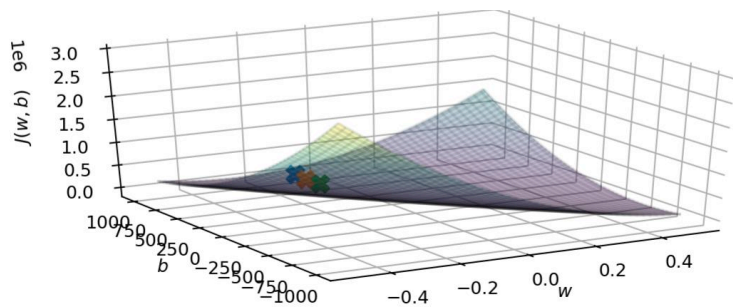
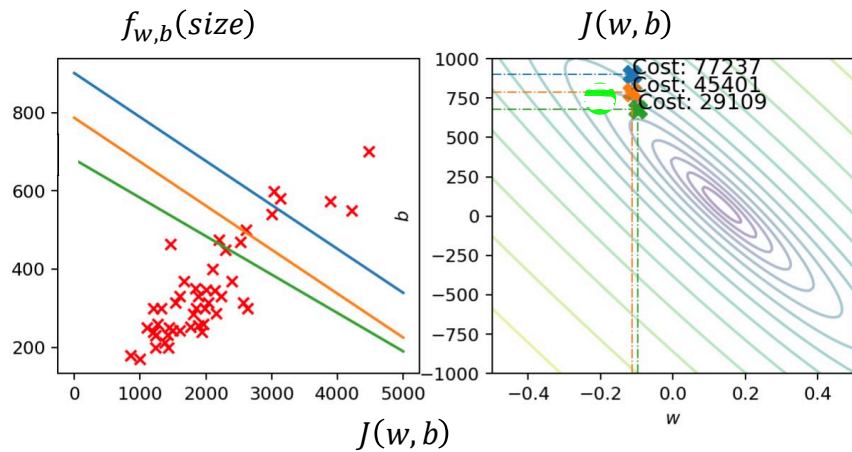
Running
Gradient Descent

price in
\$1000's

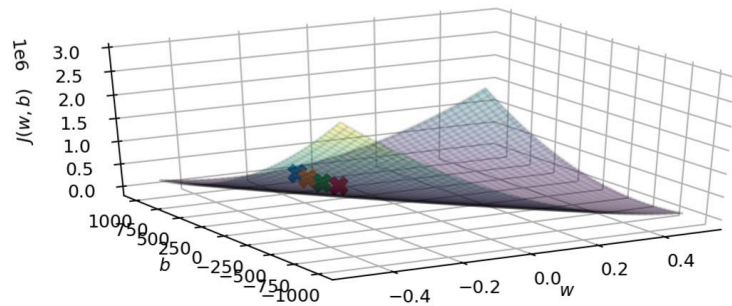
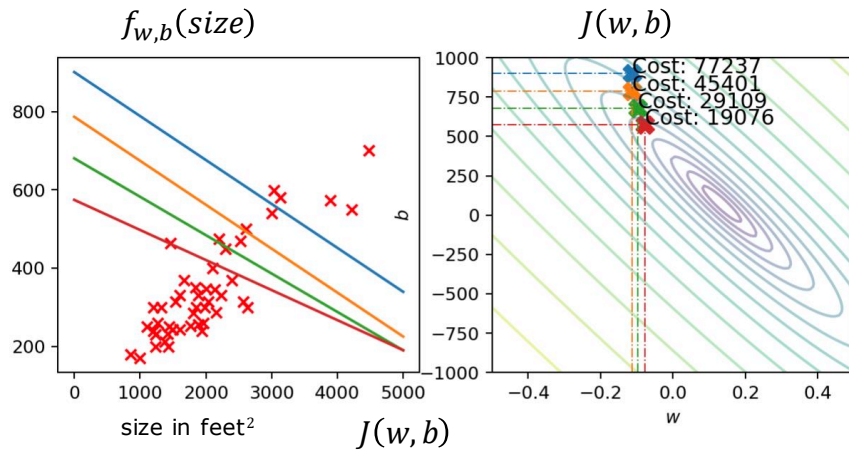


price in
\$1000's

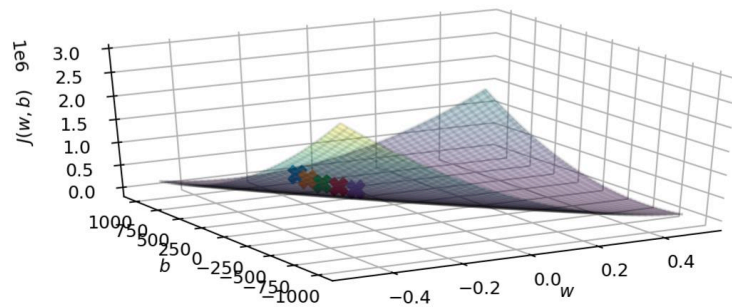
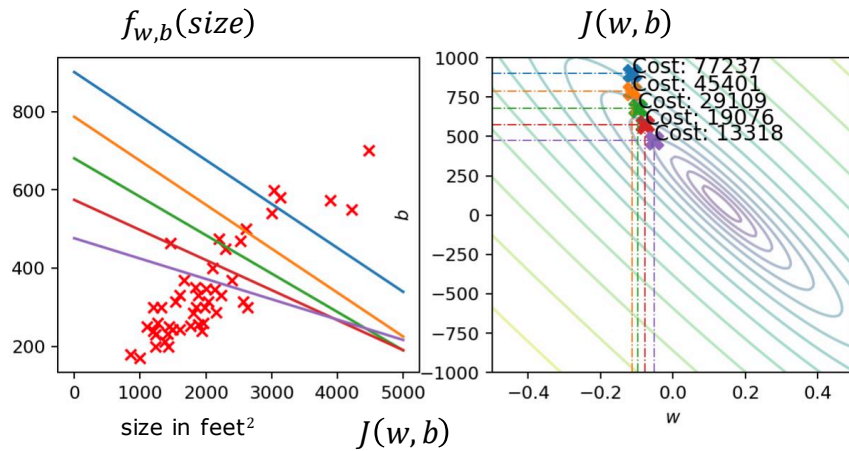




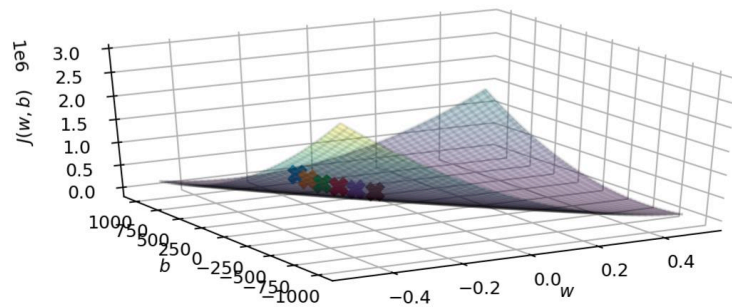
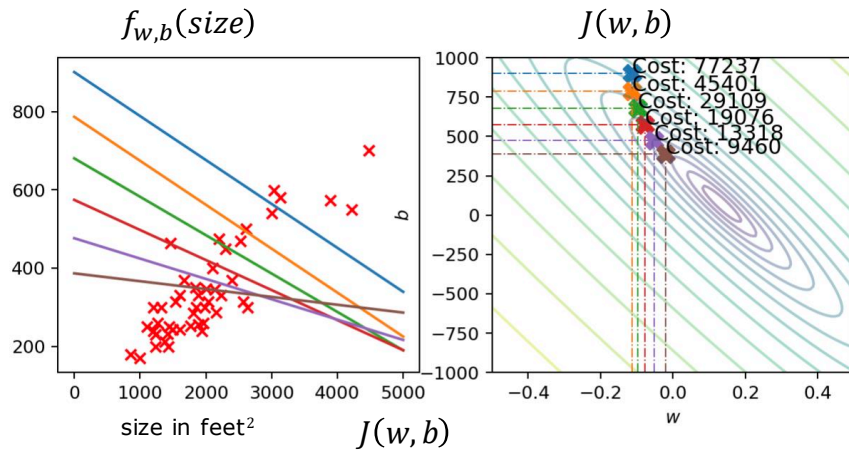
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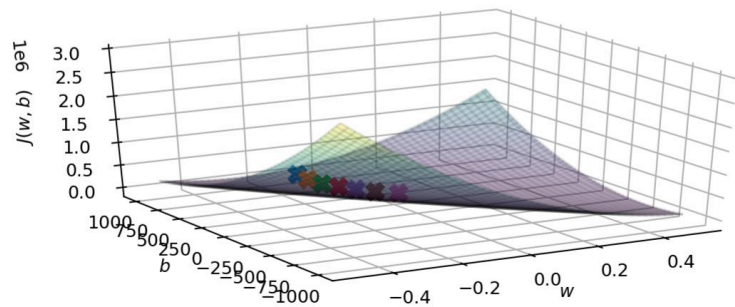
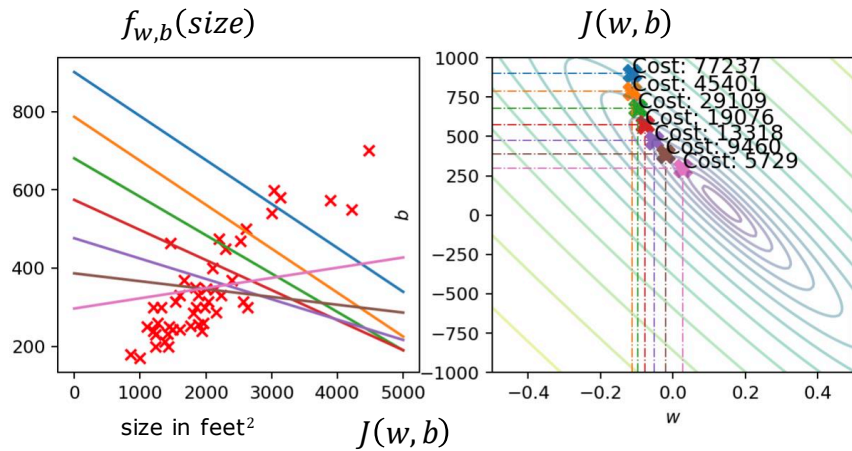
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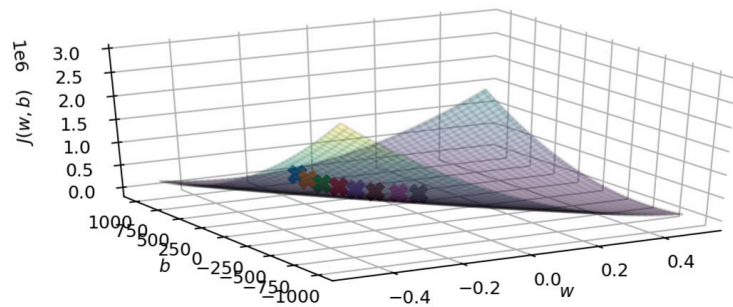
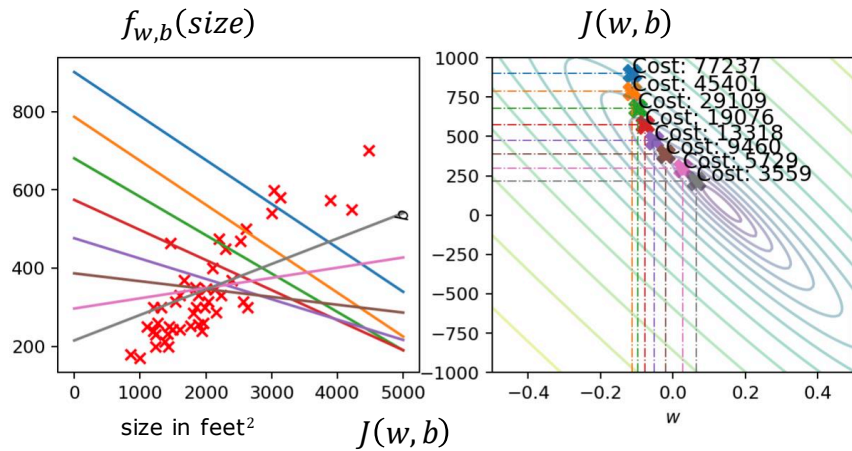
price in
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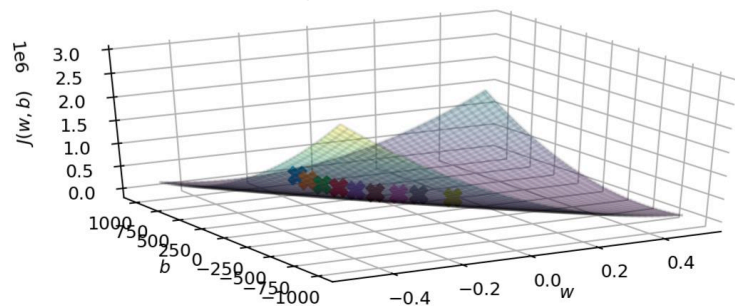
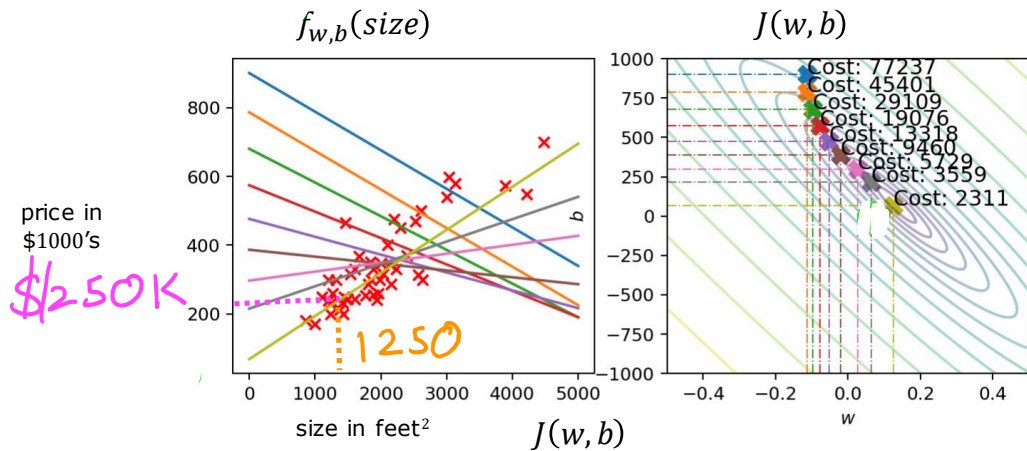


price in
\$1000's



price in
\$1000's





"Batch" gradient descent

THE BATCH

"Batch": Each step of gradient descent uses all the training examples.

other gradient descent: subsets

